

Threat Intelligence & IOCs

SOC Analyst Internship — Week 5, Task 14

Intern	Yusra Imran
Task	Week 5 - Task 14: Threat Intelligence & IOCs
Tools Used	TryHackMe, AbuseIPDB, VirusTotal
Deadline	September 19, 2026, 10:00 PM (Pakistan Time)

Objective

Understand the basics of threat intelligence and Indicators of Compromise (IOCs), and use industry-standard lookup tools (AbuseIPDB, VirusTotal) to investigate a suspicious IP address and a file hash, concluding with a verdict for each.

1. IOC Types

An Indicator of Compromise (IOC) is a piece of forensic data that signals a system may have been breached or targeted by an attacker. The four common IOC types are:

IOC Type	Example
IP Address	185.220.101.45
Domain	devret.xyz
URL	hxxp://devret[.]xyz/4833mt...
File Hash	275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f (SHA256)

2. TryHackMe Rooms Completed

Both compulsory rooms for this task were completed in full (100%):

- **Threat Intelligence Tools** — OSINT tools for security threat assessments (9/9 tasks)
- **Intro to Cyber Threat Intel** — CTI fundamentals, lifecycle, standards and frameworks (5/5 tasks)

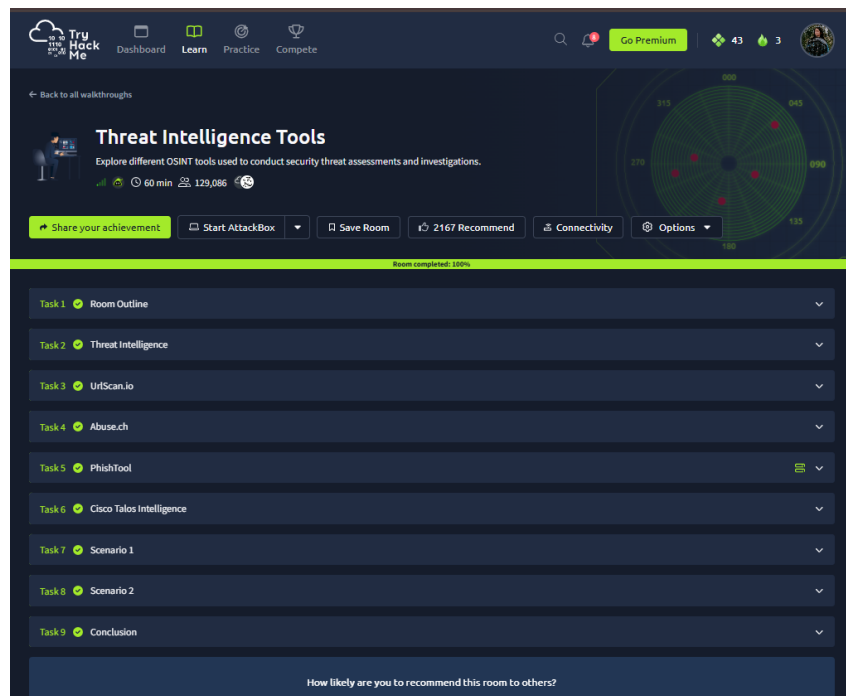


Figure 1: TryHackMe – Threat Intelligence Tools room, 100% completed

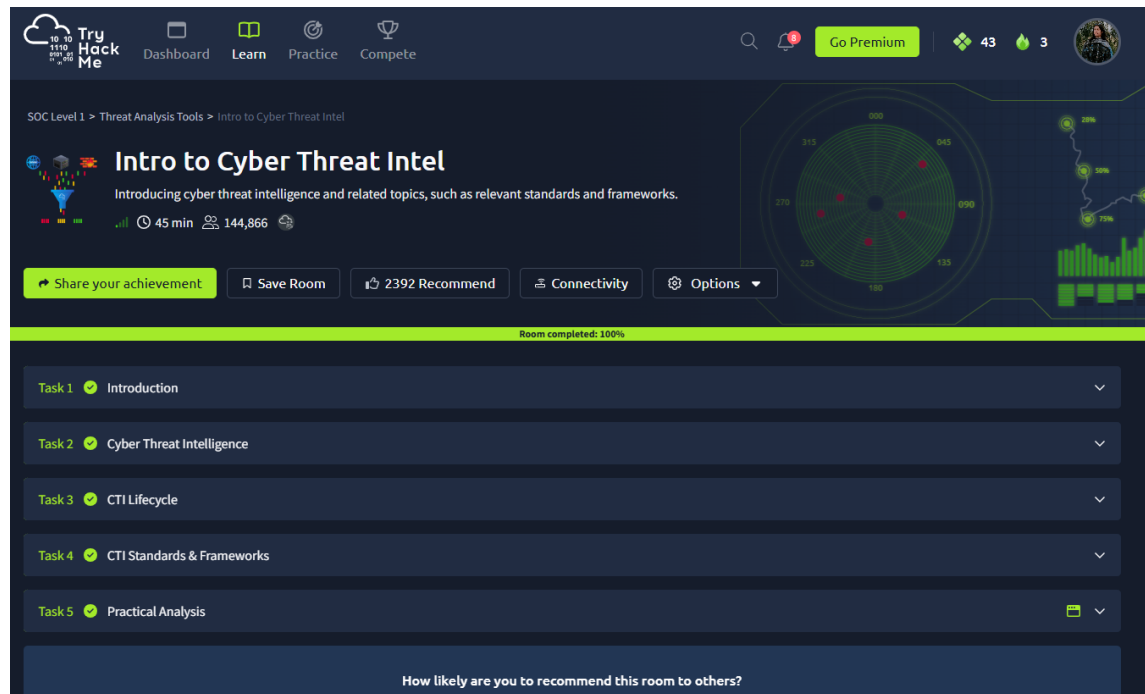


Figure 2: TryHackMe – Intro to Cyber Threat Intel room, 100% completed

3. Suspicious IP Investigation (AbuseIPDB)

The IP address 185.220.101.45 was checked on AbuseIPDB for its reputation and abuse history.

AbuseIPDB » 185.220.101.45

Check an IP Address, Domain Name, Subnet, ASN, or IP List
e.g. 119.73.119.196, microsoft.com, 5.188.10.0/24, AS15169, or [Multi-Search](#) **NEW**

185.220.101.45 **CHECK**

185.220.101.45 was found in our database!

This IP was reported 6,860 times. Confidence of Abuse is 100%. ?

100%

This address is a Tor exit node. Neither the owner nor the provider are directly behind the offending action.

ISP	Network for Tor-Exit traffic.
Usage Type	University/College/School
ASN	AS60729
Hostname(s)	tor-exit-45.for-privacy.net
Domain Name	for-privacy.net
Country	Germany
City	Berlin, State of Berlin

IP info including ISP, Usage Type, and Location provided by IPInfo. Updated weekly.

[REPORT 185.220.101.45](#) [WHOIS 185.220.101.45](#)

IP Abuse Reports for 185.220.101.45

This IP address has been reported a total of 6,860 times from 683 distinct sources. 185.220.101.45 was first reported on December 3rd 2020, and the most recent report was 3 hours ago.

Recent Reports: We have received reports of abusive activity from this IP address within the last week. It is potentially still actively engaged in abusive activities.

Figure 3: AbuseIPDB report for 185.220.101.45

Findings:

Field	Value
Abuse Confidence Score	100%
Total Reports	6,860 reports from 683 distinct sources
First Reported	December 3, 2020
Most Recent Report	3 hours ago (still active)
ISP / Usage Type	Network for Tor-Exit traffic / University-College-School
Hostname	tor-exit-45.for-privacy.net
Location	Berlin, Germany

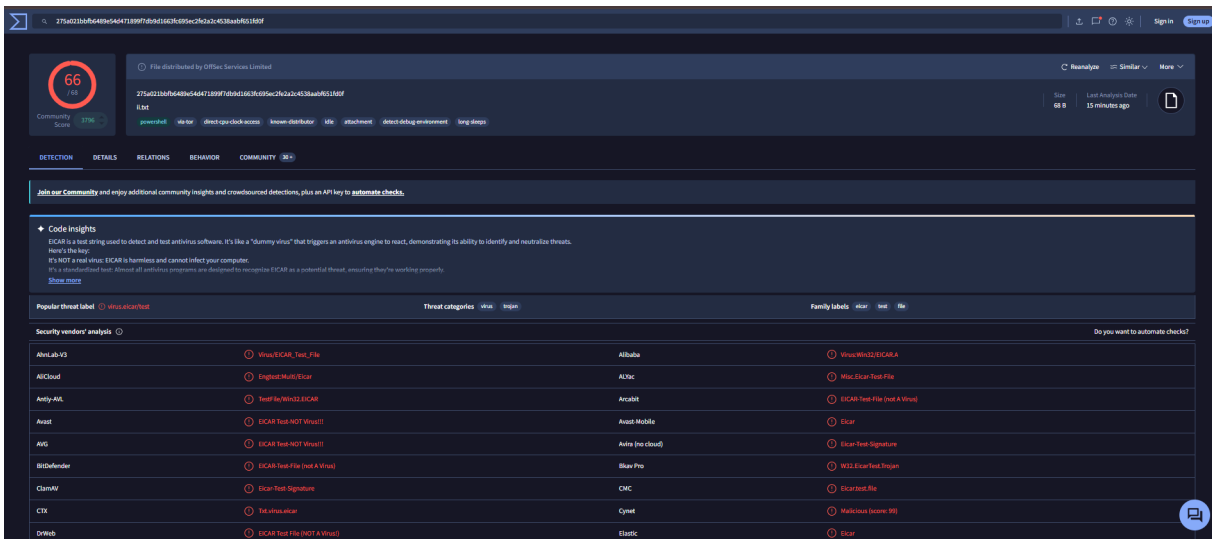
IP Verdict:

MALICIOUS / HIGH-RISK

The IP operates as a Tor exit node, which anonymizes the true source of traffic; AbuseIPDB itself notes the owner/provider is not directly behind the offending action. However, a 100% abuse confidence score, 6,860 cumulative reports from 683 independent sources, and activity as recent as 3 hours before this check indicate a consistent, ongoing pattern of abusive traffic (scanning, brute-force attempts, etc.) originating from this exit node. In a SOC context, this IP should be flagged/blocked as high-risk regardless of the underlying anonymization layer.

4. File Hash Investigation (VirusTotal)

The SHA256 hash 275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f was checked on VirusTotal.



Findings:

Field	Value
Detection Score	66/68 security vendors flagged the file
File Name	ii.txt
File Size	68 bytes
Popular Threat Label	virus.eicar/test
Threat Categories	Virus, Trojan
Sample Vendor Labels	"EICAR Test-NOT Virus!!!" (Avast), "EICAR-Test-File (not a Virus)" (BitDefender), "EICAR Test File (NOT A Virus!)" (DrWeb)

Hash Verdict:

DETECTED BUT NON-MALICIOUS (TEST FILE)

Although 66 of 68 vendors flagged the hash under virus/trojan threat categories, the vendor labels themselves consistently identify it as the **EICAR Anti-Virus Test File** — a standardized, harmless string created by the European Institute for Computer Antivirus Research specifically to test whether antivirus software is functioning correctly. It cannot infect a system. This case illustrates an important SOC analysis principle: a high detection count alone is not sufficient for a malicious verdict — the actual vendor labels/context must be reviewed to distinguish test artifacts from genuine malware.

5. Summary

- Completed both compulsory TryHackMe rooms: Threat Intelligence Tools and Intro to Cyber Threat Intel (100% each).
- Listed the 4 core IOC types: IP address, domain, URL, and file hash.
- Investigated IP 185.220.101.45 on AbuseIPDB: 100% abuse confidence, 6,860 reports, active as recently as 3 hours prior - verdict: Malicious/High-Risk (Tor exit node with sustained abuse pattern).
- Investigated a SHA256 file hash on VirusTotal: 66/68 detections, identified as the EICAR test file - verdict: Detected but Non-Malicious.
- Demonstrated the SOC analyst practice of correlating detection counts with vendor-label context before finalizing a verdict.